

12 Further processes and techniques	12.1	Strategies, techniques and approaches to explore, create and evaluate design ideas: a) user-centred design: ○ framework process ○ problem solving ○ user needs, wants and values ○ limitations of end user consideration b) circular economy – biologically-based systems and an understanding of how waste and pollution can be eliminated c) systems thinking – the influence of systems on commercial activity to enable all elements of a manufacturing enterprise to work together.
	12.2	Applications, characteristics, advantages and disadvantages of the following project management strategies: a) critical path analysis – the handling of complex and time sensitive operations b) scrum – how flexible, holistic product development is achieved c) Six Sigma – the improvement of output quality of a process by identifying and removing the causes of defects and setting value targets of: ○ reduce process cycle time ○ reduce pollution ○ reduce costs ○ increase customer satisfaction ○ increase profits.
	12.3	The cost, sales, profit and market implications to the designer and manufacturer of the stages of a product's life cycle: ● Introduction Stage ● Growth Stage ● Maturity Stage ● Decline Stage.